

Directed Acyclic Graphs

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Outline

- Why DAGs appear
- Topological sorting
- Kahn's algorithm
- DFS viewpoint
- Dynamic programming on DAGs
- Problem patterns



Why another graph family?

- Some directed graphs describe **dependencies**: one task must finish before another can start
- Examples: course prerequisites, build systems, recipe steps, project schedules
- In these problems, cycles are not just inconvenient—they make the plan impossible
- DAGs are exactly the directed graphs where dependency order makes sense

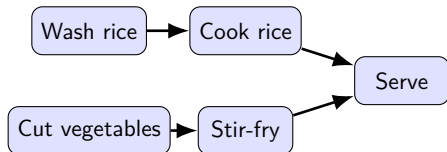


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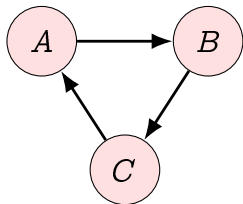
Why DAGs appear

A dependency graph



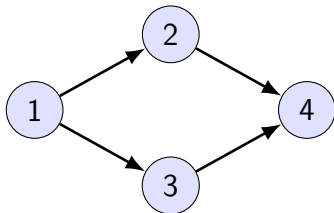
- Vertices are tasks
- Edge $u \rightarrow v$: do u before v
- Several valid schedules may exist

What goes wrong with a cycle?



- *A* waits for *C*
- *C* waits for *B*
- *B* waits for *A*
- No task can be first

DAG = directed + acyclic



- **Directed:** edges have a direction
- **Acyclic:** no directed cycle exists
- Every edge points “forward” in some valid order

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Topological sorting

Topological sorting

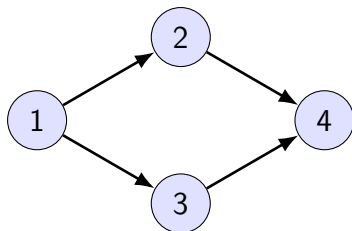
Definition

A **topological order** of a directed graph is an ordering of the vertices such that every edge $u \rightarrow v$ places u before v .

- A **topological sort** is the process of finding one such order
- It is a valid schedule for all dependencies
- A directed graph has a topological order **if and only if** it is acyclic
- The order is often not unique

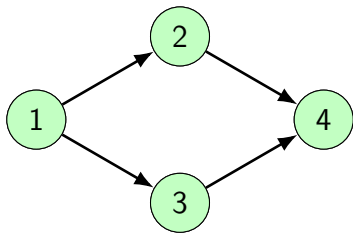


One graph, many valid orders



- Valid: 1, 2, 3, 4
- Also valid: 1, 3, 2, 4
- Invalid: 2, 1, 3, 4
because $1 \rightarrow 2$

Reading a topological order



- Start with a vertex that has no unmet prerequisite
- After placing 1, vertex 2 becomes available
- Vertex 3 is also available
- Only then can 4 appear

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Kahn's algorithm

The indegree idea

- The **indegree** of a vertex is how many incoming edges it has
- A vertex with indegree 0 has no unmet prerequisite
- So it can safely be placed next in a topological order
- After removing it, some neighbors may become indegree 0

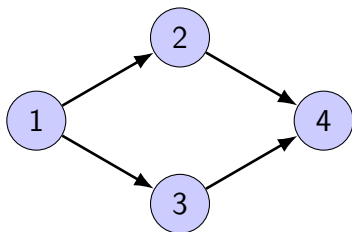


Kahn's algorithm

- Compute every indegree
- Put all indegree-0 vertices into a queue
- Repeatedly pop one vertex, append it to the order, and remove its outgoing edges
- Whenever a neighbor's indegree becomes 0, push it into the queue
- If fewer than n vertices are processed, a cycle exists



Kahn trace



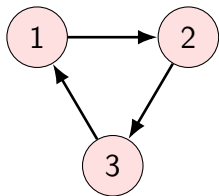
- Initial indegrees:
0, 1, 1, 2
- Queue sequence:
 $[1] \rightarrow [2, 3] \rightarrow$
 $[3] \rightarrow [4]$
- One valid order:
1, 2, 3, 4

Kahn code

```
1  vector<int> indeg(n + 1, 0);
2  for (int u = 1; u <= n; u++) {
3      for (int v : adj[u]) indeg[v]++;
4  }
5
6  queue<int> q;
7  for (int u = 1; u <= n; u++) {
8      if (indeg[u] == 0) q.push(u);
9  }
10 vector<int> order;
11 while (!q.empty()) {
12     int u = q.front();
13     q.pop();
14     order.push_back(u);
15
16     for (int v : adj[u]) {
17         if (--indeg[v] == 0) q.push(v);
18     }
19 }
20 if ((int)order.size() != n) {
21     // A cycle exists: no topological order was found.
22 }
```



Detecting a cycle



- Every vertex has indegree 1
- Initial queue is empty
- Processed count is $0 < n$
- Therefore a directed cycle exists

Example: Course Schedule

CSES: Course Schedule

Given course prerequisites, output one valid order or report that it is impossible.

- Vertices are courses
- Edge $a \rightarrow b$: take a before b
- Kahn's algorithm builds a schedule
- If the order length is smaller than n , output IMPOSSIBLE



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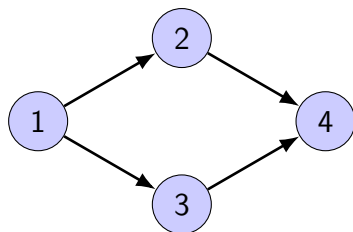


DFS viewpoint

DFS gives another order

- In DFS, a vertex finishes only after all reachable descendants are finished
- Therefore, if we record vertices on exit, descendants appear before ancestors
- Reversing that finish order gives a topological order
- A back edge to an active vertex reveals a directed cycle

DFS finish-order trace



- Finish order: 4, 2, 3, 1
- Reverse it
- Topological order:
1, 3, 2, 4

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Dynamic programming on DAGs

Why DAGs unlock DP

- DP needs subproblems to be solved before larger problems use them
- A topological order gives exactly that dependency order
- Process vertices from left to right in topological order
- Each edge relaxes information forward once its source is already known

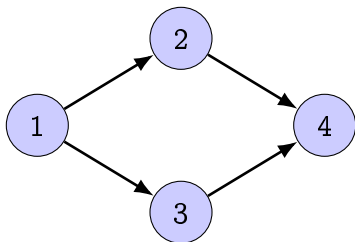


Longest path in a DAG

- In general graphs, longest simple path is difficult because avoiding cycles couples choices
- In a DAG, cycles do not exist
- So we can compute longest paths with dynamic programming
- If $u \rightarrow v$, then $dp[v] = \max(dp[v], dp[u] + 1)$



DAG DP trace



- Topological order:
1, 2, 3, 4
- DP values: 0, 1, 1, 2
- Each edge pushes
information forward

Longest-path DP code

```
1  const int NEG_INF = -1e9;
2  vector<int> dp(n + 1, NEG_INF);
3  dp[start] = 0;
4
5  for (int u : topo) {
6      if (dp[u] == NEG_INF) continue;
7      for (int v : adj[u]) {
8          dp[v] = max(dp[v], dp[u] + 1);
9      }
10 }
```



Example: Longest Flight Route

CSES: Longest Flight Route

Find a route from city 1 to city n with as many cities as possible in a DAG.

- Use a topological order
- Let $dp[v]$ be the maximum number of edges on a route to v
- Store $parent[v]$ when improving $dp[v]$
- Reconstruct the chosen route at the end

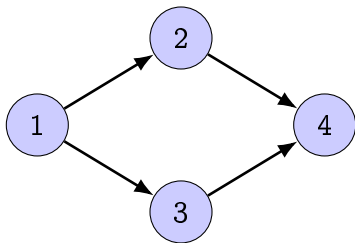


Counting paths in a DAG

- Sometimes we want the **number** of valid routes, not just one best route
- Let $ways[v]$ be the number of paths from the start to v
- Process vertices in topological order
- For every edge $u \rightarrow v$, add $ways[u]$ into $ways[v]$
- In code: $ways[v] += ways[u]$



Counting-paths trace



- Topological order:
1, 2, 3, 4
- Path counts:
1, 1, 1, 2
- Incoming routes add together

Counting-paths code

```
1  const int MOD = 1e9 + 7;
2  vector<int> ways(n + 1, 0);
3  ways[start] = 1;
4
5  for (int u : topo) {
6      for (int v : adj[u]) {
7          ways[v] = (ways[v] + ways[u]) % MOD;
8      }
9  }
```



Example: Game Routes

CSES: Game Routes

Count how many routes lead from level 1 to level n in a directed acyclic graph.

- Let $\text{ways}[v]$ be the number of routes reaching v
- Start with $\text{ways}[1] = 1$
- Push counts forward in topological order
- Take answers modulo $10^9 + 7$



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Problem patterns

Recognizing DAG problems

- The input describes prerequisites, precedence, or one-way progress
- You need a valid ordering, or need to decide whether one exists
- Values should flow from earlier states to later states
- The graph has no reason to revisit a state once progress is made



Which tool should I reach for?

Need	Use	Why
Build any valid order	Topological sort	Respect dependencies
Detect impossible schedule	Kahn or DFS coloring	Cycles block ordering
Longest / count / combine paths	DAG DP	Topological order solves subproblems first
Lexicographically smallest order	Min-heap Kahn	Choose the smallest ready vertex



Common mistakes

- Using BFS order and assuming it is topological order
- Forgetting that many valid topological orders may exist
- Running DAG DP before confirming the graph is acyclic
- Initializing DP incorrectly for unreachable vertices



The End

Thanks for listening.

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